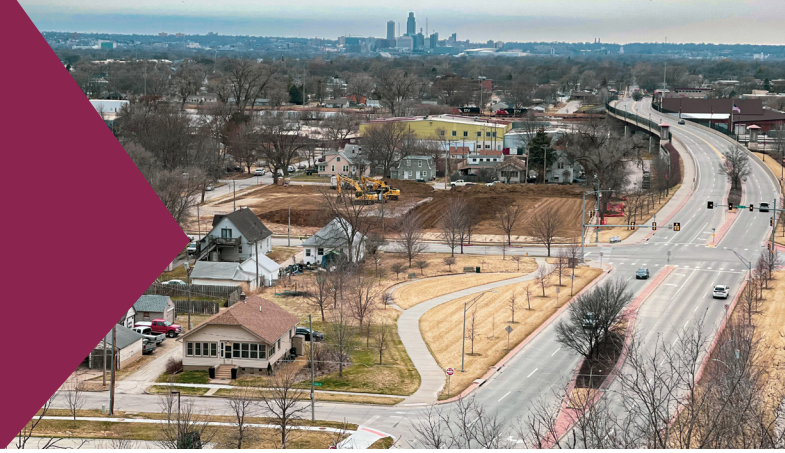
U.S. Department of Transportation
ITS Joint Program Office

Machine Learning-Enhanced Roadway Asset Identification and Classification
Nebraska DOT



Source: Getty Images, Inc.

Challenges

Transportation asset management (TAM) activities at the Nebraska Department of Transportation (NDOT) support the maintenance of reliable inventory and condition information to support infrastructure stewardship, maintenance planning, and long-range investment decisions. The agency's latest transportation asset management plan (TAMP) emphasized condition assessment, lifecycle planning, risk management, and performance-based decisionmaking for roadway infrastructure assets. Effective asset management programs rely on comprehensive and reliable infrastructure inventories to support maintenance strategies and long-term financial planning. NDOT sought to inventory previously uninventoried safety-critical assets using existing data sources.

Existing practices relied heavily on manual inventories and periodic field reviews, which made it difficult to maintain up-to-date, systemwide asset records for many types of assets. Some assets have not been inventoried and have unknown characteristics and conditions without manual review.

Agency Background

Based on its 2022 TAMP, NDOT maintains a mature statewide asset management culture. The agency focuses on data-driven lifecycle management, performance targets, and transportation system preservation. The 2022 NDOT TAMP emphasizes strategic asset management, performance monitoring, inspection methods, and data-driven prioritization. The agency maintains extensive inspection and inventory programs that depend on large quantities of roadway and bridge condition data. For example, NDOT stores bridge inspection information in centralized databases and uses it to inform maintenance strategies and lifecycle decisions.

The TAMP documents NDOT's broader organizational emphasis on infrastructure preservation, freight mobility, operational efficiency, and long-term system stewardship. The plan also shows that NDOT leadership was accustomed to using formal prioritization frameworks, lifecycle-cost analysis, and performance-management methods to support investment decisions. NDOT's machine learning (ML) pilot activities thus emerged from an agency culture emphasizing innovation, safety improvement, TAM, and performance-based infrastructure stewardship. Agency leadership at first approached ML cautiously. However, they ultimately supported the low-risk guardrail and marked pedestrian crossing pilots to automate data collection and reduce operational costs.

At a Glance

AGENCY: **Nebraska Department of Transportation**

USE CASE AREA: **Asset management**

MATURITY LEVEL: **Pilot project**

METHOD: **Machine learning; convolutional neural network (CNN) object detection and classification**

DATA SOURCE: **Geocoded imagery from statewide roadway network profiling (approximately 2.5 million images)**

BUDGET: **$25,000 per pilot project**

KEY OUTCOME: **Statewide guardrail and endcap inventory plus classification of more than 1,300 marked crosswalks, with agency review of all model outputs**

Method Used

This effort represents a decision-support framework. NDOT explored how ML image processing could support data collection and inventory maintenance activities. NDOT used ML methods to identify and classify roadway assets automatically using pre-existing geocoded imagery from statewide roadway network profiling activities. The deployment focused on detecting and classifying guardrails, guardrail attenuators, and marked pedestrian crossings. For these particular pilot projects, CNNs were the chosen ML method for object detection. The broader national research identified computer vision, roadway imagery analysis, and automated asset extraction as emerging transportation applications for ML and deep learning methods.

The ML applications supported asset inventory development by extracting roadway asset information from image data, eliminating the need for staff to review every individual image manually. The resulting analytical outputs supported asset management workflows by providing structured roadway asset information that agency personnel could review and incorporate into existing TAM and safety improvement activities. The pilot deployment functioned as a decision-support tool rather than an automated decisionmaking system.

Development Timeline

NDOT's ML pilot activities, reported on in 2024 by the National Cooperative Highway Research Program (NCHRP), leveraged an image database from the agency's 2019 roadway-network profiling to create a statewide inventory. These activities occurred within a national context in which agencies were increasingly exploring ML and computer-vision applications for infrastructure-inventory and condition-assessment activities.

Development Process

NDOT explored using ML to detect guardrails and classify them by type. A consulting firm labeled approximately 1,500 pavement-profiler van images to train the ML model. The van is outfitted with multiple cameras that take roughly 15 pictures per second. After developing and training the ML model, the firm then applied the model to approximately 2.5 million roadway images to detect guardrails and create a statewide inventory. The firm then trained two object classification models: one to classify the guardrails, and a second to classify endcaps for those guardrails classified as Type 1.

Figure 1 shows a typical image from NDOT's pavement-profiler vans, and Figure 2 shows a photo that is zoomed in on the guardrail. Software-generated bounding boxes show the guardrail and endcaps that ML has identified. The learning objective of these photos is to show how vans, cameras, and ML models can provide efficient asset management data for real-world guardrails and endcaps.



Source: NDOT.

Figure 1. Photo. Typical profiler van view image.

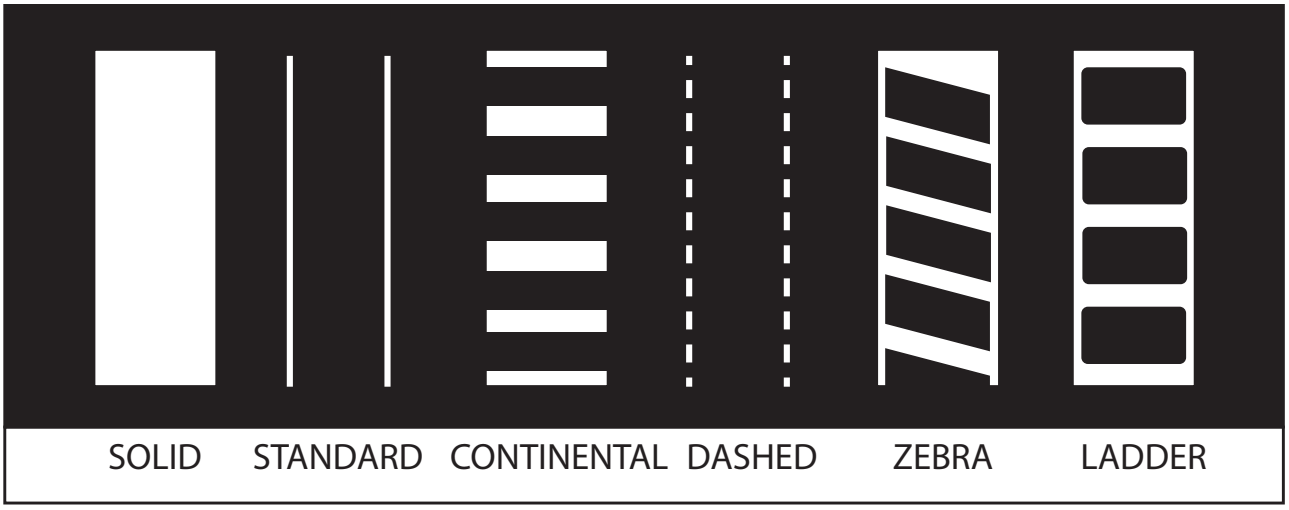


Source: NDOT.

Figure 2. Photo. Example guardrail image processed by the object identification model.

NDOT also explored using ML to detect marked pedestrian crossings. Following a first pass of ML object detection, human analysts inspected the ML outputs and manually assigned crosswalk marking types (i.e., solid, standard, continental, dashed, zebra, and ladder). This two-step process resulted in a classification of more than 1,300 marked crosswalks across the State's roadways. In both the guardrail and crosswalk pilot projects, CNNs were the preferred form of ML used for object detection.

Figure 3 illustrates the six recognized crosswalk-marking types, and Figure 4 shows a photo containing two detected crosswalks. The photo contains vehicles and objects on either side of the road, as well as crosswalks that the ML model may typically detect and classify. Software-generated bounding boxes show the identified crosswalks.



Source: NDOT.

Figure 3. Illustration. Six recognized crosswalk types.



Source: NDOT.

Figure 4. Photo. Example crosswalk image processed by the object identification model.

The broader NCHRP research described how agencies increasingly explored ML methods to automate portions of roadway image review and infrastructure inventory activities. Agencies conducted these explorations while continuing to rely on human oversight and engineering review. Consistent with this pattern, NDOT personnel had the opportunity to review and interpret analytical outputs generated by the pilot applications before applying that information to TAM and safety improvement activities.

Key Design Choices

- Narrowly scoped roadway asset inventory pilot applications
- Use of existing roadway profiling imagery and geocoded image datasets
- Application of computer vision and CNN-based asset detection methods
- Human review of analytical outputs before operational use

Resources

Both projects (guardrail detection, marked pedestrian crossing detection) were very cost effective for NDOT, with each having a budget of $25,000. Part of this cost-effectiveness came from capitalizing on existing image data. In addition, both projects had limited, well-defined scopes with clear goals that helped simplify execution. The large quantity of roadway images (2.5 million) created substantial data-storage and -transfer needs. NDOT transferred roadway imagery files using physical hard drives because the size of the datasets created practical bandwidth and data-transfer limitations. The deployment also relied on consultant experts in ML.



Source: Getty Images, Inc.

Key Implementation Factors

NDOT focused on narrowly scoped pilot applications using existing roadway imagery collected during operations related to roadway-network profiling. The first application detected guardrails, classified the guardrails, and classified the guardrail attenuators. The second identified and classified marked pedestrian crossings. By leveraging existing roadway imagery datasets, NDOT explored ML applications without establishing entirely new field-data collection programs. ML methods supported roadway asset detection and classification activities, while agency personnel reviewed and interpreted analytical outputs before incorporating information into TAM and safety improvement workflows. This human-centered implementation approach aligned with NCHRP findings describing how agencies often introduced ML through limited pilot deployments that supplemented rather than replaced existing engineering and asset management practices. Within this national context, NDOT's pilot activities showed how computer vision and ML methods could support development of roadway asset inventories using large roadway-imagery datasets collected during statewide roadway-profiling operations.

Outputs, Outcomes, and Impacts

NDOT's pilot ML applications generated roadway asset inventory information derived from geocoded roadway imagery collected during activities related to statewide roadway-network profiling. During the ML-assisted development of NDOT's inventories, analysts reviewed interim outputs such as classification accuracy percentages. There were three guardrail classification outputs in the developed inventory: crash cushion, Type 1, and Type 2 attenuator. For the Type 1 guardrails, there were five endcap classification outputs. The six crosswalk types shown in figure 3 were another output of the AI system.

Key Outputs

- Roadway asset inventory information derived from roadway imagery
- Identified locations of guardrails, guardrail attenuators, and marked pedestrian crossings
- Asset classification distinguishing guardrail types and attenuator subtypes
- Structured roadway asset information supporting TAM and safety improvement activities
- Roadway-imagery and machine learning models for identifying real-world crosswalks and efficiently generating asset management data

NDOT's pilot ML applications showed how computer vision and ML methods could improve the efficiency of roadway image review, asset identification, and inventory maintenance activities. These automated methods operated successfully while maintaining human oversight and engineering review processes. For example, ML-based guardrail detection and endcap classification were particularly beneficial due to the need for an up-to-date inventory to replace endcaps from specific manufacturers. Moreover, the guardrail and endcap classifications facilitated creation of a geographic information system layer for locating relevant guardrail information as points and line features on a map. The broader NCHRP research described how agencies increasingly explored ML methods to support infrastructure-inventory development, roadway-imagery analysis, and TAM activities using existing transportation datasets and operational workflows.



Source: Getty Images, Inc.

Development Challenges

The large size of the roadway imagery datasets created data storage and transfer challenges that ultimately resulted in shipment of physical hard drives for external processing because of bandwidth and file-size limitations. The pilot applications also faced limitations associated with roadway imagery coverage and asset visibility. Some roadway assets were difficult to identify consistently because roadway imagery collected from profiler vehicles did not always capture assets located outside the primary roadway corridor or outside the camera's field of view. The NCHRP research also documented that agencies often relied on external technical experts when exploring early ML pilot applications because many agencies had limited in-house experience with computer vision and ML methods.

Lessons Learned

- **Start with narrowly scoped pilot applications.** NDOT's pilot activities focused on specific roadway asset inventory tasks, including guardrail detection, guardrail attenuator classification, and marked pedestrian crossing identification using roadway imagery. The NCHRP research described how agencies often introduced ML through limited pilots that addressed well-defined operational needs while limiting implementation risk.
- **Leverage existing roadway imagery and operational datasets.** NDOT applied ML methods to roadway imagery already collected during activities related to statewide roadway-network profiling rather than establishing entirely new field-data-collection programs. The pilot deployment showed how existing roadway imagery datasets could support exploratory ML applications for roadway asset inventory development.
- **Human review and validation are important.** The pilot applications showed that roadway asset detections derived from ML methods would still benefit from human review and interpretation before incorporation into TAM and safety improvement workflows. Factors such as asset visibility, roadway geometry, image quality, and camera field of view limited the ability of automated methods to identify every roadway asset consistently.
- **External technical experts may support early pilot deployments.** The NCHRP research documented that transportation agencies often relied on external support resources and technical experts during early ML pilot activities because many agencies had limited in-house experience with computer vision and ML methods.

Transferability

NDOT's experience shows that ML adoption in asset management can be effective when agencies begin with targeted, data-ready applications that deliver clear value. The pilots applied careful scoping, reuse of existing imagery, human oversight of automated outputs, and incremental confidence building among leadership and staff. These lessons imply that early ML deployments can prioritize feasibility, transparency, and learning over scale. These lessons may be transferable to agencies seeking to evaluate asset inventories and foundational data quality using ML while preserving established inspection practices and decision authority.

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